

Pavan Reddy | AI/ML Engineer

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SUMMARY

AI/ML Engineer with 5+ years of experience building machine learning solutions, recommendation systems, predictive analytics models, semantic search applications, and Retrieval-Augmented Generation (RAG) platforms across retail, healthcare, and financial services domains. Skilled in Python, SQL, PySpark, Apache Spark, Databricks, MLflow, Airflow, AWS, Azure, and Large Language Models (LLMs). Proven experience improving forecasting accuracy, search relevance, deployment efficiency, and operational performance through scalable machine learning pipelines, distributed data processing, MLOps practices, and production-ready AI solutions.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Scala, Java, R, Bash

Machine Learning & Statistics: Classification, Regression, Clustering, Time Series Forecasting, Predictive Modeling, Recommendation Systems, Anomaly Detection, Feature Engineering, Hyperparameter Optimization, Statistical Analysis, Model Evaluation, A/B Testing

Deep Learning: PyTorch, TensorFlow, Keras, Neural Networks, CNNs, RNNs, LSTMs, Transformer Architectures

Generative AI & NLP: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Semantic Search, Vector Search, Embeddings, Prompt Engineering, LangChain, LangGraph, LlamaIndex, OpenAI API, Text Classification, Information Extraction, Sentiment Analysis, Document Understanding

Data Engineering & Distributed Computing: PySpark, Apache Spark, Databricks, Delta Lake, Apache Airflow, ETL Development, Data Transformation, Workflow Orchestration, Distributed Data Processing

MLOps & Software Engineering: MLflow, Docker, Kubernetes, CI/CD Pipelines, GitHub Actions, Jenkins, Model Deployment, Model Monitoring, Experiment Tracking, Model Validation, Version Control, REST APIs

Cloud Platforms: AWS SageMaker, Amazon S3, AWS Lambda, AWS Glue, Amazon Redshift, Azure Machine Learning, Azure Data Factory, Azure Synapse Analytics, Microsoft Azure

Databases & Storage: PostgreSQL, MySQL, SQL Server, Snowflake, MongoDB, Redis, Pinecone, ChromaDB

Data Analysis & Visualization: Pandas, NumPy, Power BI, Tableau, Matplotlib, Plotly

Tools & Methodologies: Git, GitHub, Azure DevOps, Jira, Confluence, Agile, Scrum

PROFESSIONAL EXPERIENCE

AI/ML Engineer

Mar 2025 - Present | USA

Walmart Global Tech

- Delivered a Retrieval-Augmented Generation platform integrating enterprise knowledge repositories, vector search, and large language models, reducing information retrieval time by 67% while improving response quality across business operations.
- Increased forecasting accuracy by 22% through feature engineering, model selection, and iterative optimization techniques applied to inventory planning, demand forecasting, and operational analytics initiatives.
- Reduced model deployment timelines by 75% through automated machine learning workflows utilizing Databricks, MLflow, Airflow, and cloud-native orchestration processes supporting production environments.
- Improved semantic search precision by 34% through embedding optimization, retrieval tuning, and vector indexing strategies that enhanced discovery of critical business information.
- Accelerated distributed data processing workloads by 41% through PySpark optimization, partition management, and efficient transformation logic supporting large-scale machine learning pipelines.
- Improved recommendation quality by 18% through behavioral feature development, ranking enhancements, and continuous experimentation supporting customer engagement and personalization initiatives.
- Reduced cloud infrastructure costs by 26% through workload optimization, resource allocation improvements, and scheduling enhancements across model training and inference environments.
- Cut manual document review efforts by 58% through natural language processing solutions supporting classification, extraction, and content analysis across business workflows.
- Maintained 99.5% pipeline reliability by implementing automated validation controls, monitoring frameworks, and proactive alerting mechanisms across machine learning operations.
- Partnered with product, engineering, and business teams to deliver AI-driven capabilities that improved adoption rates, reduced response times, and supported operational decision-making.
- Shortened model evaluation cycles by 45% through reusable experimentation frameworks, standardized testing approaches, and automated performance reporting capabilities.

- Developed reusable machine learning components adopted across multiple initiatives, reducing development effort and accelerating solution delivery across engineering teams.

AI/ML Engineer
Accenture

Apr 2020 - Dec 2023 | India

- Developed machine learning solutions supporting healthcare, retail, and financial services engagements, improving forecasting accuracy, customer targeting effectiveness, and operational planning outcomes.
- Improved churn prediction accuracy by 24% through feature engineering, data enrichment, and model optimization techniques applied to customer retention initiatives.
- Reduced data preparation effort by 52% through scalable ETL workflows and automated preprocessing pipelines supporting machine learning and analytics initiatives.
- Increased anomaly detection precision by 29% through refined feature selection strategies, statistical modeling techniques, and performance validation approaches.
- Improved document classification accuracy by 31% through transformer-based natural language processing models supporting content management and workflow automation initiatives.
- Reduced model training times by 38% through distributed computing techniques, Spark optimization, and efficient data partitioning strategies.
- Improved recommendation relevance by 21% through ranking model enhancements, user behavior analysis, and personalization techniques supporting digital platform experiences.
- Reduced manual reporting effort by 60% through automated analytics workflows, scheduled processing pipelines, and model-driven insight generation capabilities.
- Increased stakeholder adoption of machine learning solutions through explainability frameworks, transparent evaluation methodologies, and performance measurement standards.
- Supported migration of analytics and machine learning workloads to AWS and Azure environments, improving scalability, deployment flexibility, and operational efficiency.
- Reduced production issues by 35% through monitoring enhancements, validation controls, root-cause analysis, and continuous model performance improvements.
- Created reusable machine learning assets, implementation templates, and development standards that improved consistency and accelerated delivery across client engagements.

EDUCATION

Computer Science (Cyber Security)
Webster University

Dec 2025 | MO, USA